

Topological semimetals carrying arbitrary Hopf numbers: Fermi surface topologies of a Hopf link, Solomon's knot, trefoil knot, and other linked nodal varieties

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We propose a type of Hopf semimetal indexed by a pair of numbers (p, q) , where the Hopf number is given by pq . The Fermi surface is given by a preimage of the Hopf map, which consists of loops nontrivially linked for a nonzero Hopf number. The Fermi surface forms a torus link, whose examples are a Hopf link indexed by $(1, 1)$, Solomon's knot $(2, 1)$, a double Hopf link $(2, 2)$, and a double trefoil knot $(3, 2)$. We may choose p or q to be a half integer, where the Fermi surface is a torus knot, such as a trefoil knot $(3/2, 1)$. It is even possible to make the Hopf number an arbitrary rational number, where a semimetal whose Fermi surface forms open strings is generated.

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Introduction. Weyl semimetals are described by a two-band model equipped with a point node in three-dimensional (3D) space [1,2]. It is characterized by a monopole charge in momentum space [3]. Line-nodal semimetals or loop-nodal semimetals are also possible in 3D space, where the zero-energy Fermi surface is given by a line or a closed loop [5–15]. Recently, a nodal-chain semimetal was proposed, where loop nodes touch [4]. A natural question is whether nontrivial Fermi surfaces made of loop nodes such as links and knots are possible.

The two-band Hamiltonian $H = \mathbf{S}(\mathbf{k}) \cdot \boldsymbol{\sigma}$ is a prototype of Hopf insulators [16–21], where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices and $\mathbf{S}(\mathbf{k})$ is the pseudospin field. By considering a normalized pseudospin $\mathbf{n}(\mathbf{k}) = \mathbf{S}(\mathbf{k})/|\mathbf{S}(\mathbf{k})|$, we can construct the sphere surface $\mathbb{S}^2$. On the other hand, the 3D Brillouin zone is identical to the torus $\mathbb{T}^3$. A homotopy from $\mathbb{T}^3$ to $\mathbb{S}^2$ is characterized by the Hopf number. It has been argued that Hopf insulators with arbitrary Hopf numbers are possible [17]. Nontrivial Hopf textures are also discussed in cold atoms [22,23], light fields [24], and liquid crystal [25].

In this Rapid Communication we investigate topological semimetals, where Fermi surfaces consist of nontrivial loops with arbitrary Hopf numbers. We explore the Hamiltonian $H = S_x \sigma_x + S_z \sigma_z$, where the zero-energy condition reads $n_y = \pm 1$. The Fermi surface is a preimage of the points $\mathbf{n}_{\pm} = (0, \pm 1, 0)$ in the mapping $\mathbb{T}^3 \rightarrow \mathbb{S}^2$, and it consists of two loops. They are linked for a nonzero Hopf number. We construct Fermi surfaces comprising a Hopf link, Solomon's knot, and others: See Fig. 1. They are “torus links” lying on the surface of a torus. Furthermore, we construct Fermi surfaces comprising torus knots by choosing half-integer Hopf numbers, among which there arises in particular a trefoil-node Fermi surface. We can even choose any one rational number as a Hopf number, where the Fermi surface forms open strings, though it no longer describes a topological semimetal.

Torus-link semimetals. The Hamiltonian of topological Hopf insulators is given by [16–21]

$$H(\mathbf{k}) = \mathbf{S}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad (1)$$

where $\mathbf{S}(\mathbf{k})$ is the pseudospin field. It is defined in terms of two complex fields as $\mathbf{S} = (\eta_{\uparrow}^p, \eta_{\downarrow}^q) \boldsymbol{\sigma} (\eta_{\uparrow}^p, \eta_{\downarrow}^q)^T$, where $\eta_{\uparrow}$ and $\eta_{\downarrow}$ are

complex numbers given by [16–20]

$$\begin{aligned} \eta_{\uparrow}(\mathbf{k}) &= \sin k_x + i \sin k_y, \\ \eta_{\downarrow}(\mathbf{k}) &= \sin k_z + i(\cos k_x + \cos k_y + \cos k_z - m), \end{aligned} \quad (2)$$

while p and q are integers. We note that p and q are originally introduced as a set of coprime integers [17], but here we do not impose that. We will see that p and q can be generalized even to rational numbers, though a cut is introduced. It is convenient to introduce the $\mathbb{CP}^1$ field $\mathbf{z}$ and the normalized pseudospin $\mathbf{n}$,

$$\mathbf{z} \equiv \begin{pmatrix} z_{\uparrow} \\ z_{\downarrow} \end{pmatrix} = \frac{1}{\sqrt{|\eta_{\uparrow}|^2 + |\eta_{\downarrow}|^2}} \begin{pmatrix} \eta_{\uparrow}^p \\ \eta_{\downarrow}^q \end{pmatrix}, \quad (3)$$

$$\mathbf{n}(\mathbf{k}) = \mathbf{S}(\mathbf{k})/|\mathbf{S}(\mathbf{k})| = \mathbf{z}^{\dagger} \boldsymbol{\sigma} \mathbf{z}, \quad (4)$$

where $\mathbf{n}(\mathbf{k})$ is explicitly represented as [16–20]

$$n_x + i n_y = \frac{2\eta_{\uparrow}^p \bar{\eta}_{\downarrow}^q}{|\eta_{\uparrow}|^{2p} + |\eta_{\downarrow}|^{2q}}, \quad n_z = \frac{|\eta_{\uparrow}|^{2p} - |\eta_{\downarrow}|^{2q}}{|\eta_{\uparrow}|^{2p} + |\eta_{\downarrow}|^{2q}}. \quad (5)$$

We consider a class of pseudospin textures indexed by a pair of numbers (p, q) in the Hamiltonian (1).

The $\mathbb{CP}^1$ field takes values on the 3D sphere $\mathbb{S}^3$ since it contains four real numbers, $N_1 = \text{Re } z_{\uparrow}(\mathbf{k})$, $N_2 = \text{Im } z_{\uparrow}(\mathbf{k})$, $N_3 = \text{Re } z_{\downarrow}(\mathbf{k})$, and $N_4 = \text{Im } z_{\downarrow}(\mathbf{k})$, together with the normalization condition $\sum_i N_i^2 = 1$. It gives a mapping $\mathbb{T}^3 \rightarrow \mathbb{S}^3$, and for the Brillouin zone it is a 3D torus. On the other hand, the normalized pseudospin expressed as $\mathbf{n} = \mathbf{z}^{\dagger} \boldsymbol{\sigma} \mathbf{z}$ defines a mapping $\mathbb{S}^3 \rightarrow \mathbb{S}^2$, because it takes values on the sphere $\mathbb{S}^2$. Consequently, the underlying structure of the Hamiltonian (1) is a mapping $\mathbb{T}^3 \rightarrow \mathbb{S}^3 \rightarrow \mathbb{S}^2$ from the Brillouin zone to the pseudospin space.

The combined mapping $\mathbb{T}^3 \rightarrow \mathbb{S}^2$ is indexed by the Hopf number [16–21]

$$\Gamma(p, q) = -\frac{1}{2\pi^2} \int_{\text{BZ}} d\mathbf{k} \varepsilon_{\mu\nu\rho\tau} N_{\mu} \partial_{k_x} N_{\nu} \partial_{k_y} N_{\rho} \partial_{k_z} N_{\tau}. \quad (6)$$

By using (2) and (3) in this formula, we obtain

$$\Gamma(p, q) = \begin{cases} 0, & |m| > 3, \\ pq, & 1 < |m| < 3, \\ -2pq, & |m| < 1. \end{cases} \quad (7)$$

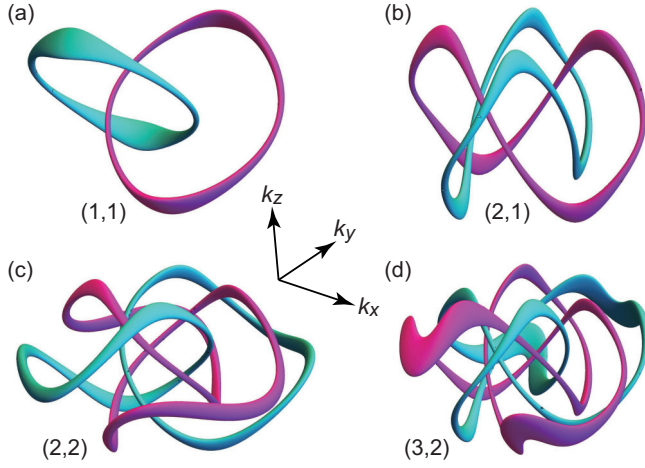


FIG. 1. Bird's-eye view of the almost zero-energy surface of the Hamiltonian H_{xz} . (a) Hopf-link semimetal with $(p,q) = (1,1)$, (b) Solomon's-knot semimetal with $(2,1)$, (c) double Hopf-link semimetal with $(2,2)$, which consists of two Hopf links. (d) Double trefoil-knot semimetal with $(3,2)$. We have chosen $m = 2$ in Eq. (2) to draw the figures. The preimage of $n_y = 1$ is colored in magenta, while that of $n_y = -1$ is colored in cyan.

We note that the definition of the Hopf number (6) is different from the previous literature [17], in which it is defined in terms of $(\eta_\uparrow, \eta_\downarrow)$ rather than $(z_\uparrow, z_\downarrow)$. The formula (7) is understood intuitively as follows: When $p = q = 1$, it is a well-known formula for the Hopf number [17], which indicates that there exists one circle in the interior of the torus and one circle around its axis of rotational symmetry. Now, $z_\uparrow \propto \eta_\uparrow^p$ and $z_\downarrow \propto \eta_\downarrow^q$ imply that there exist p and q of these circles.

The energy spectrum of the Hamiltonian (1) reads $E(\mathbf{k}) = \pm\sqrt{S_x^2(\mathbf{k}) + S_y^2(\mathbf{k}) + S_z^2(\mathbf{k})}$. The Fermi surface of the topological Hopf insulator is constructed by the intersection of three curved surfaces, $S_x = S_y = S_z = 0$. In general, the intersection of the three surfaces is null, which results in an insulating state.

In order to construct a model having a Fermi surface, it is necessary to reduce the number of the conditions on the zero-energy states. There are three trivial ways to do so. We may employ any one of the conditions, $S_x = S_z = 0$, $S_y = S_z = 0$, or $S_x = S_y = 0$. Obviously, they are the zero-energy conditions of the following three Hamiltonians,

$$H_{xz}(\mathbf{k}) = S_x(\mathbf{k})\sigma_x + S_z(\mathbf{k})\sigma_z, \quad (8)$$

$$H_{yz}(\mathbf{k}) = S_y(\mathbf{k})\sigma_y + S_z(\mathbf{k})\sigma_z, \quad (9)$$

$$H_{xy}(\mathbf{k}) = S_x(\mathbf{k})\sigma_x + S_y(\mathbf{k})\sigma_y, \quad (10)$$

respectively. The Fermi surface constructed by the intersection of two curved surfaces is a line node in general. In all these models the pseudospin field is defined by $\mathbf{S} = (|\eta_\uparrow|^{2p} + |\eta_\downarrow|^{2q})\mathbf{n}$, with $\mathbf{n} = \mathbf{z}^\dagger \boldsymbol{\sigma} \mathbf{z}$ based on the same CP^1 field (3). Only the Hamiltonian H_{xz} preserves the combination symmetry PT of the time reversal T and inversion symmetry P .

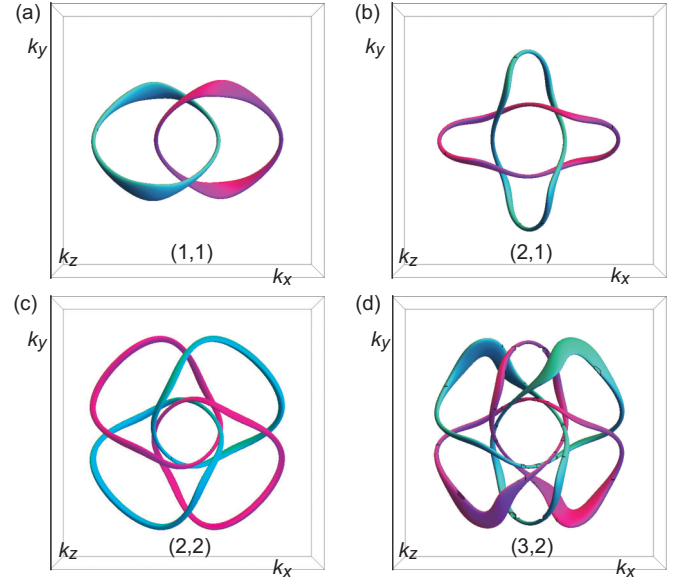


FIG. 2. Top view of almost zero-energy surface of the Hamiltonian H_{xz} in the 3D Brillouin zone. See the caption of Fig. 1 for (a)–(d).

The H_{xz} model. First, we investigate the Hamiltonian H_{xz} . From the normalization condition on $\mathbf{n}(\mathbf{k})$, the zero-energy condition is expressed as $n_y(\mathbf{k}) = \pm 1$. Namely, the Fermi surface is a preimage of the points $\mathbf{n}_\pm = (0, \pm 1, 0)$ in a combined Hopf map $\mathbb{T}^3 \rightarrow \mathbb{S}^2$, which is a circle $\mathbb{S}^1$ in the 3D Brillouin zone. Consequently, there are at least two loops corresponding to the preimage of $n_y(\mathbf{k}) = 1$ and $n_y(\mathbf{k}) = -1$ in the 3D Brillouin zone. These two loops form a link when the Hopf number is nonzero.

Various links indexed by a pair (p,q) are realized in the 3D Brillouin zone. We show an almost zero-energy surface $E = \delta$ with $|\delta| \ll 1$ for (p,q) in Figs. 1 and 2. The preimage of $n_y(\mathbf{k}) = 1$ is colored in magenta and that of $n_y(\mathbf{k}) = -1$ is colored in cyan. For example, the Hopf link and Solomon's knot are realized by taking pairs $(1,1)$ and $(2,1)$, respectively. The Fermi surface for $(3,2)$ is given by the combination of two trefoils, which we call a double trefoil.

We recall the terminology in link theory. A link lying on the surface of a torus is called a torus link. The torus link $T(p,q)$ winds q times around a circle in the interior of the torus, and p times around its axis of the rotational symmetry. In the present context, the surface determined by the condition $S_z = 0$ gives a torus. Thus the node indexed by (p,q) is the torus link $T(2p,2q)$, where the factor 2 appears because it is a preimage of the two points $S_\pm = (0, \pm 1, 0)$. According to link theory, $T(2p,2q)$ is identical to $T(2q,2p)$. Furthermore, the $T(-2p,2q)$ link and $T(2p,-2q)$ are mirror images of $T(2p,2q)$.

If p and q are not relatively prime, we have a torus link with more than one component. The number of the loops is given by $\text{gcd}(2p,2q) = 2 \text{gcd}(p,q)$, where gcd represents the greatest common divisor. For example, the Fermi surface consists of four loop nodes for $p = 2$ and $q = 2$. This looks a bit odd since we have discussed that two loops arise as a preimage of $n_y = \pm 1$. How such a Fermi surface consisting of more

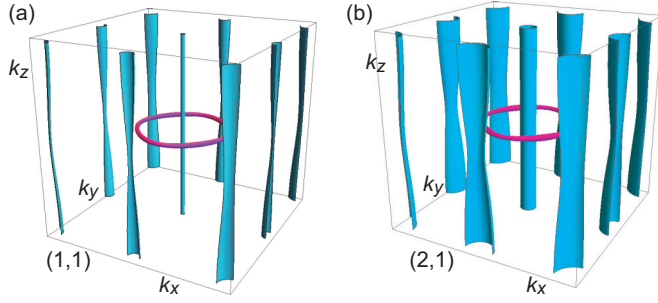


FIG. 3. Bird's-eye view of the almost zero-energy surface of the Hamiltonian H_{xy} for (a) the pair $(p, q) = (1, 1)$ and (b) $(2, 1)$.

than two loops is realized for $\gcd(p, q) \neq 1$ is an interesting problem. We assume $\gcd(p, q) = s$. Then we can write $p = sp'$ and $q = sq'$, where p' and q' are coprime integers. The solutions are given by $\eta_{\uparrow}^{1/s}$ and $\eta_{\downarrow}^{1/s}$, where $\eta_{\uparrow}$ and $\eta_{\downarrow}$ satisfy $n_y = \pm 1$ for the model with p' and q' . The roots of $\eta_{\uparrow}^{1/s}$ and $\eta_{\downarrow}^{1/s}$ have s solutions, which result in s Fermi loops.

In the following, we derive the equations to determine the link. The zero-energy states must satisfy the condition $n_z = \mathbf{z}^\dagger \sigma_z \mathbf{z} = |z_{\uparrow}|^2 - |z_{\downarrow}|^2 = 0$. By combining it with the normalization condition $|z_{\uparrow}|^2 + |z_{\downarrow}|^2 = 1$, the CP^1 field is parametrized as

$$z_{\uparrow} = \frac{1}{\sqrt{2}} e^{i\theta_{\uparrow}}, \quad z_{\downarrow} = \frac{1}{\sqrt{2}} e^{i\theta_{\downarrow}}. \quad (11)$$

The zero-energy states correspond to $n_y = \pm 1$, which reads

$$n_y = \mathbf{z}^\dagger \sigma_y \mathbf{z} = -\sin(\theta_{\uparrow} - \theta_{\downarrow}) = \pm 1.$$

The solution is given by $\theta_{\uparrow} - \theta_{\downarrow} = \mp \frac{\pi}{2}$. We find the relation $z_{\downarrow} = \pm i z_{\uparrow}$, or

$$\eta_{\downarrow}^q = \pm i \eta_{\uparrow}^p. \quad (12)$$

These two equations determine the Fermi surface made of links.

The H_{yz} model. The Fermi surface of H_{yz} is topologically equivalent to that of the H_{xz} model. The only difference is that the Fermi surface is rotated 90° between the H_{xz} and H_{yz} models.

The H_{xy} model. The Fermi surface of H_{xy} looks very different, where some of the Fermi surfaces form lines penetrating the whole Brillouin zone: See Fig. 3. The model H_{xy} is instructive since we obtain various analytical expressions.

The zero-energy condition in the model H_{xy} is given by the preimage of $n_z = |z_{\uparrow}|^2 - |z_{\downarrow}|^2 = \pm 1$. By combining it with the normalization condition $|z_{\uparrow}|^2 + |z_{\downarrow}|^2 = 1$, the preimage of $n_z = 1$ is given by $|z_{\uparrow}|^2 = 1$ and $|z_{\downarrow}|^2 = 0$, which is equivalent to the condition $|z_{\downarrow}|^2 = |\eta_{\downarrow}|^{2q} = 0$, where

$$|\eta_{\downarrow}|^2 = \sin^2 k_z + (\cos k_x + \cos k_y + \cos k_z - m)^2 = 0. \quad (13)$$

The solution is

$$\cos k_x + \cos k_y = m - 1, \quad k_z = 0, \quad (14)$$

which represents q -fold degenerate loop nodes.

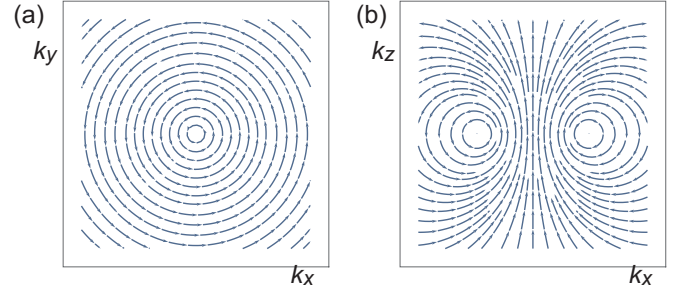


FIG. 4. (a) Stream plot of the Berry curvature (A_x, A_y) along the k_x - k_y plane with $k_z = 0$. (b) Stream plot of the Berry curvature (A_x, A_z) on the k_x - k_z plane at $k_y = 0$.

On the other hand, the preimage of $n_z = -1$ is given by $|z_{\uparrow}|^2 = 0$ and $|z_{\downarrow}|^2 = 1$, which is equivalent to the condition $|z_{\uparrow}|^2 = |\eta_{\uparrow}|^{2p} = 0$, where

$$|\eta_{\uparrow}|^2 = \sin^2 k_x + \sin^2 k_y = 0.$$

The solution represents p -fold degenerate line nodes along the k_z direction described by $k_x = n_x \pi$, $k_y = n_y \pi$, k_z with $-\pi \leq k_z \leq \pi$, where n_x and n_y are integers.

We may also analyze this Hamiltonian system from the viewpoint of the Berry curvature. We introduce a continuum model defined by

$$\begin{aligned} \eta_{\uparrow}(\mathbf{k}) &= k_x + i k_y, \\ \eta_{\downarrow}(\mathbf{k}) &= k_z + i [3 - m - (k_x^2 + k_y^2 + k_z^2)/2]. \end{aligned} \quad (15)$$

The Fermi surface is topologically equivalent to the original model (2). The only difference is the approximation of the loop (14) by the circle $k_x^2 + k_y^2 = 2(3 - m)$, which does not change the linking structure.

By using them we may derive explicitly the eigenstate $|\psi\rangle$ of the Hamiltonian (10). The Berry connection $A_{k_i} = -i \langle \psi | \partial_{k_i} | \psi \rangle$ is given by

$$A_k = \frac{2kqk_z}{k_z^4 + k_z^2(k^2 + 2m - 4) + (k^2 + 2m - 6)^2}, \quad (16)$$

$$A_\theta = \frac{p}{2}, \quad (17)$$

$$A_z = \frac{q(k_z^2 - k^2 - 2m + 6)}{k_z^4 + k_z^2(k^2 + 2m - 4) + (k^2 + 2m - 6)^2}, \quad (18)$$

where we have introduced the polar coordinate $k_x = k \cos \theta$ and $k_y = k \sin \theta$. We show the Berry curvature in Fig. 4. We find a vortex structure along a line node (cyan line in Fig. 3) described by $k_x = 0$, $k_y = 0$, k_z with $-\pi \leq k_z \leq \pi$, and a circle described by $k_x^2 + k_y^2 = 6 - 2m$, $k_z = 0$. Actually, this line node has a p -fold degeneracy. Indeed, we calculate the Berry phase along a loop encircling the line node to find that

$$\Gamma_B = \oint A_i dk_i = \int_0^{2\pi} A_\theta d\theta = p\pi, \quad (19)$$

which is quantized and represents the degeneracy. In the same way, the Berry phase along a small circle around a loop node

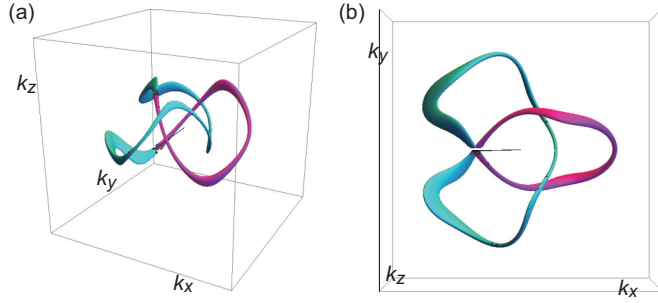


FIG. 5. Trefoil-knot semimetal with $p = 3/2$ and $q = 1$. (a) Bird's-eye view and (b) top view of the almost zero-energy surface. There is a cut on the $k_y = 0$ plane for $k_x < 0$, where the magenta and cyan curves touch.

(magenta loop in Fig. 3) is obtained as

$$\Gamma_B = \oint A_i dk_i = \int_0^{2\pi} A_\theta d\theta = q\pi, \quad (20)$$

representing the degeneracy of the loop node. Thus the indices p and q are detected separately, while only the product pq appears in the Hopf number. It is interesting that the Berry phase can be larger than π , which is not the case in unlinked nodal semimetals.

Torus-knot semimetals. We have seen that the Fermi surfaces consist of at least two loops, which form a torus link. An interesting problem is whether we can construct a Fermi surface consisting of one nontrivial loop, which is a torus knot. Since the number of loops is given by $2 \gcd(p, q)$, we must take one of the p and q noninteger. We find that a torus-knot Fermi surface is realized by taking a half-integer p . For example, we can realize a trefoil Fermi surface by taking $p = 3/2$ and $q = 1$, which is shown in Fig. 5. There is a cut in the momentum space due to the square root contribution in the Hamiltonian. Namely, only the magenta or cyan curve does not form a closed loop but forms a loop with a combination of them.

Open-string semimetals. It is possible to even choose any rational numbers for p or q , which creates an open-string Fermi

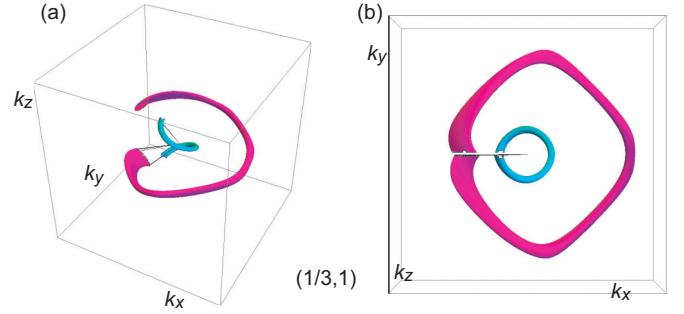


FIG. 6. Open-string semimetal with $p = 1/3$ and $q = 1$. (a) Bird's-eye view and (b) top view of the almost zero-energy surface. There is a cut along the $k_y = 0$ plane for $k_x < 0$, where the ends of the open string exist.

surface, as shown in Fig. 6. This is understood as follows. The model with p and q gives the torus link $T(2p, 2q)$. If one of the p and q is not a half integer, it cannot describe a closed loop, generating an open string. The ends of the open string are on the cut plane.

Discussion. Recently, it was pointed out [4] that chain-nodal semimetals are realized in IrF₄, where the loop nodes are not linked but are touching. By investigating similar material classes, loop-nodal semimetals with links or knots will be found. In addition, there are some proposals where topological Hopf insulators are realized in cold atoms with spin-dependent hoppings, magnetic compounds, and three-dimensional quantum walks [16, 17, 19, 26]. The model is a good starting point for further studies on linked nodal semimetals.

Note added. Recently, we became aware of closely related works [26–28], where various linked nodal semimetals are proposed. Especially, this Rapid Communication has turned out to be a generalization of Ref. [27], where only the case with $p = q = 1$ is studied.

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- [1] P. Hosur and X. L. Qi, *C. R. Phys.* **14**, 857 (2013).
 - [2] S. Jia, S.-Y. Xu, and M. Z. Hasan, *Nat. Mater.* **15**, 1140 (2016).
 - [3] S. Murakami, *New J. Phys.* **9**, 356 (2007).
 - [4] T. Bzduiek, Q.-S. Wu, A. Ruegg, M. Sigrist, and A. A. Soluyanov, *Nature (London)* **538**, 75 (2016).
 - [5] S. Mandal and N. Surendran, *Phys. Rev. B* **79**, 024426 (2009).
 - [6] A. A. Burkov, M. D. Hook, and L. Balents, *Phys. Rev. B* **84**, 235126 (2011).
 - [7] M. Phillips and V. Aji, *Phys. Rev. B* **90**, 115111 (2014).
 - [8] C. Fang, Y. Chen, H.-Y. Kee, and L. Fu, *Phys. Rev. B* **92**, 081201 (2015).
 - [9] L. S. Xie, L. M. Schoop, E. M. Seibel, Q. D. Gibson, W. Xie, and R. J. Cava, *APL Mater.* **3**, 083602 (2015).
 - [10] R. Yu, H. Weng, Z. Fang, X. Dai, and X. Hu, *Phys. Rev. Lett.* **115**, 036807 (2015).
 - [11] Y. Kim, B. J. Wieder, C. L. Kane, and A. M. Rappe, *Phys. Rev. Lett.* **115**, 036806 (2015).
 - [12] A. Yamakage, Y. Yamakawa, Y. Tanaka, and Y. Okamoto, *J. Phys. Soc. Jpn.* **85**, 013708 (2016).
 - [13] M. Ezawa, *Phys. Rev. Lett.* **116**, 127202 (2016).
 - [14] M. Hirayama, R. Okugawa, T. Miyake, and S. Murakami, *Nat. Commun.* **8**, 14022 (2017).
 - [15] Y.-H. Chan, C.-K. Chiu, M. Y. Chou, and A. P. Schnyder, *Phys. Rev. B* **93**, 205132 (2016).
 - [16] J. E. Moore, Y. Ran, and X.-G. Wen, *Phys. Rev. Lett.* **101**, 186805 (2008).

- [17] D.-L. Deng, S.-T. Wang, C. Shen, and L.-M. Duan, [Phys. Rev. B **88**, 201105\(R\) \(2013\)](#).
- [18] D.-L. Deng, S.-T. Wang, and L.-M. Duan, [Phys. Rev. B **89**, 075126 \(2014\)](#).
- [19] D.-L. Deng, S.-T. Wang, K. Sun, and L.-M. Duan, [arXiv:1612.01518](#).
- [20] R. Kennedy, [Phys. Rev. B **94**, 035137 \(2016\)](#).
- [21] C. Liu, F. Vafa, and C. Xu, [Phys. Rev. B **95**, 161116\(R\) \(2017\)](#).
- [22] Y. Kawaguchi, M. Nitta, and M. Ueda, [Phys. Rev. Lett. **100**, 180403 \(2008\)](#).
- [23] D. S. Hall, M. W. Ray, K. Tiurev, E. Ruokokoski, A. H. Gheorghe, and M. Mottonen, [Nat. Phys. **12**, 478 \(2016\)](#).
- [24] H. Kedia, I. Bialynicki-Birula, D. Peralta-Salas, and W. T. M. Irvine, [Phys. Rev. Lett. **111**, 150404 \(2013\)](#).
- [25] P. J. Ackerman and I. I. Smalyukh, [Phys. Rev. X **7**, 011006 \(2017\)](#).
- [26] W. Chen, H.-Z. Lu, and J.-M. Hou, [Phys. Rev. B **96**, 041102 \(2017\)](#).
- [27] Z. Yan, R. Bi, H. Shen, L. Lu, S.-C. Zhang, and Z. Wang, [Phys. Rev. B **96**, 041103 \(2017\)](#).
- [28] P.-Y. Chang and C.-H. Yee, [arXiv:1704.01948](#).